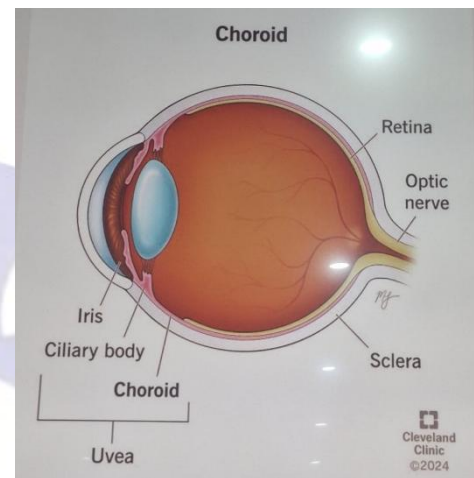
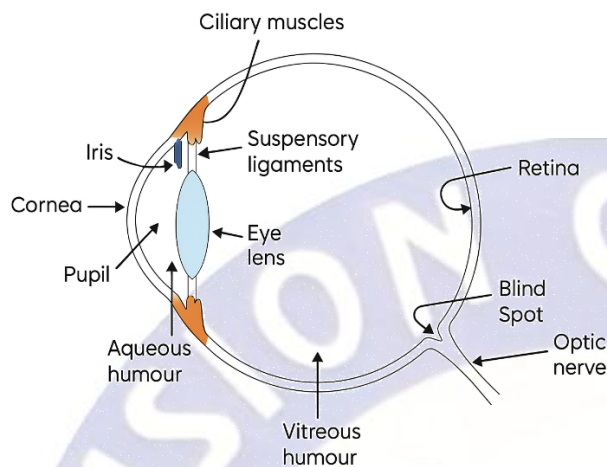


HUMAN EYES

- **The Human eyes**

- The human eyes is a natural optical device. It works on the refraction of light through a natural convex lens made of transparent living material and enables us to see things around us.



- **The important parts of human eyes are following:**

- | | |
|-------------------------------|----------------------|
| 1. Sclera. | 8. Vitreous humour. |
| 2. Cornea. | 9. Retina. |
| 3. Choroid. | 10. Optic nerve. |
| 4. Iris. | 11. Ciliary muscles. |
| 5. Pupils. | 12. Blind Spot. |
| 6. Eye lens/crystalline lens. | 13. Eye – ball. |
| 7. Aqueous humour. | |

- 1. **Sclera**

- That part of an eye which is an opaque white, protective membrane is called the sclera.
- It protects our eyes.

- 2. **Cornea**

- The front small portion of the eye made by transparent protective membrane and it is bulged out, called cornea.
- The light coming from objects enters the eye through the cornea. It is convex in shape.

- 3. **Iris**

- That part of eye which is flat, coloured, ring – shaped membrane behind the cornea.

- 4. **Choroid**

- There is a deep brown coloured surface the major structures of the eye, called choroid.

- 5. **Pupils**

- That part of an eye which is round coloured part that we can see is the iris. It has a small opening hole is called pupil, through which light enters the eye.

OR

There is a small opening hole in the middle of the iris which is called pupil of the eye.

- The pupil appears black because no light is reflected from it.

Note:

1. The pupils controls the amount of light entering the eyes and iris controls the size of pupils.

If the amount of light received by the eye is large (day), then the iris contracts the pupil (small in size) and reduces the amount of light entering the eye.

If the amount of light received by the eye is small (night), the iris expands the pupil (large in size) so, that more light may enter the eyes.

2. The adjustment of the size of the pupils takes some time.

6. Aqueous humour

- That part of the eye which is filled with a watery fluid is called the aqueous humour.
- The aqueous humour is filled between cornea and eye – lens.

Note:

Aqueous humour maintains the pressure needed to inflate the eye and provides nutrition for the central cornea and lens as they do not have blood supply.

7. Vitrous humour

- The jelly like fluid is filled between the eyes – lens and retina, called vitrous humour.
- Vitrous humour helps in maintaining the shape of the eye – ball.

8. Retina

- That part of eye which helps in bending light and the retina through on which the image is formed retina.

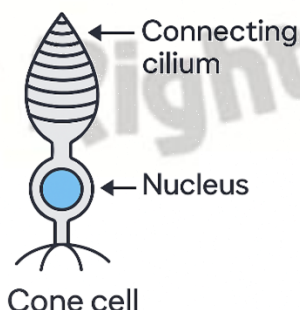
OR

The screen on which the image is formed in the eye is called retina.

- Retina has two sense receptor, they are known as rods and cones.
- Sense receptor is also called light sensitive cells.
- There are a large number of rods & cones present in the retina.
- Rods and cones respond to the intensity of light and colour of objects respectively, by generating electrical signals.

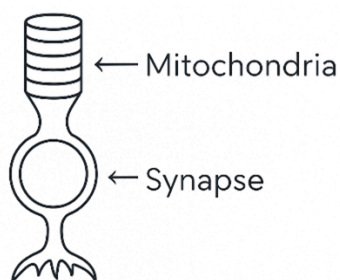
9. Cones

- The Cones are the cone – shaped cells present in the retina of an eye which are sensitive to bright/normal light.
- It causes the sensation of colour of objects in our eyes.
- It doesn't work/function in dim light.



10. Rods

- Rods are the rod – shaped cells present in the retina of an eye which are sensitive to dim light.
- It doesn't work in bright light.
- It does not provide the information about the colour of the object.



Rod Cell

▪ **Difference between rods and cones**

- There are many differences between rods and cones.

<u>Rods</u>	<u>Cones</u>
Rods are rod – shaped cells.	Cones are cone – shaped cells.
It works in dim light.	It works in bright light.
It has no information about the colour of the object.	It has the information about the colour of the object.

11. Eye lens/ crystalline lens

- The eye – lens is a convex lens made of a transparent, soft and flexible material like a jelly made of proteins being flexible, the eye – lens can change to focus light on the retina.

12. Ciliary muscles

- The flexible muscles which hold the eye lens in its position, and controls the shape of eye lens as well as its focal length (small/large).

13. Optic nerve

- The optical nerve is a bundle of one million nerve fibers that transmits visual information from the retina to the brain to help you see.

14. Blind Spot

- The blind spot is a small area of the retina insensitive to light where the optic nerve leaves the eye.

OR

The junction of optic nerve there is no light sensitive cells (rods & cones) due to which no vision is possible at that spot. This small spot is called blind spot.

15. Eye – ball

- The different parts of the eye are contained within the spherical shape is called the eye – ball.

• **Working of the eye**

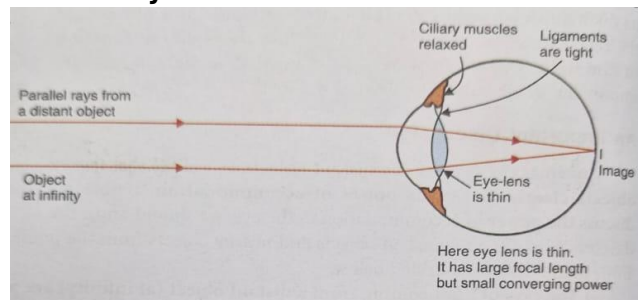
- The light rays coming from an object through the cornea of the eye, through the pupil of the eye and fall on the eye – lens. The eye – lens is a convex lens, so it converges the light rays and produces a real and inverted image of the object on the retina. Then light sensitive cells get activated and generate electrical signals. The retina sends these electrical signals to the brain through the optic nerve and gives rise to sensation of vision. Although the image formed is inverted, our mind intercepts the image as that of an erect object.

• **Accommodation**

- The ability of an eye to focus the distant object as well as the nearby objects on the retina by changing the focal length (converging power) of its lens is called accommodation.

- The Ciliary muscles adjust the focal length.

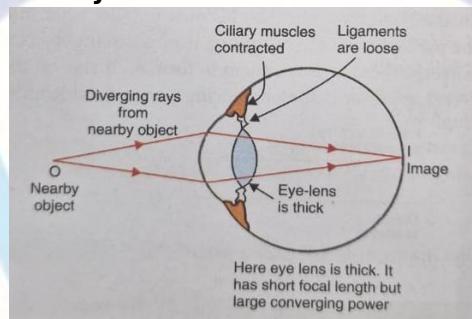
Case 1: When an object placed at infinity



Note:

- In this case, eye lens is thin so, it has large focal length but small converging power.

Case 2: When an object is placed at nearby



Note:

- In this case, eye lens is thick. So, it has short focal length but large converging power.

- **Range of vision of a normal human eye**

- The range of vision of a normal human eye is 25 cm to infinity i.e, a man able to see objects; placed between near 25 cm and far infinity.
- **Far point of normal human eye**
 - The farthest point upto which an eye can see properly is called the far point of a normal eye.
 - The far point of a normal eye is infinity (at a large distance).
- **Near point of a normal eye**
 - The closest point at which an object can be placed and see clearly is called the near point of the normal eye.
 - The near point of a normal eye is also called the least distant vision or, least distance of clear vision.
 - The standard value of the least distance of distinct vision for the normal eye is taken as 25 cm.

- **Defect of vision**

- When a human eye lose the ability to focus the image of an object on the retina, this defect is called defect of vision.
- It is also called defect of eye.

- **VISION**

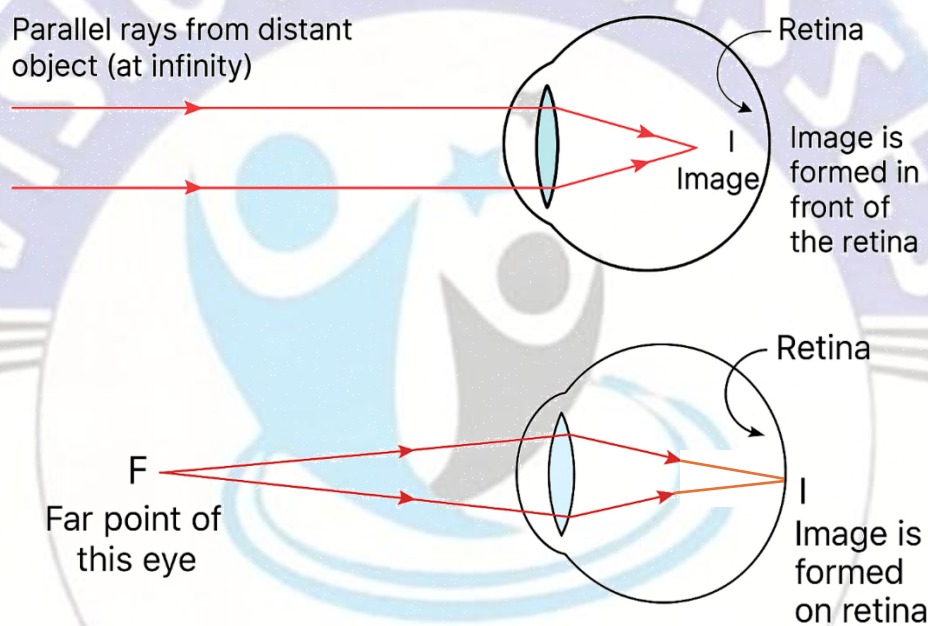
- The ability to see the objects is called vision.
- It is also called eyesight and also known as 'Drishti' in Hindi.

The common defect of vision

1. Myopia.
2. Hypermetropia.
3. Presbyopia.
4. Cataract.

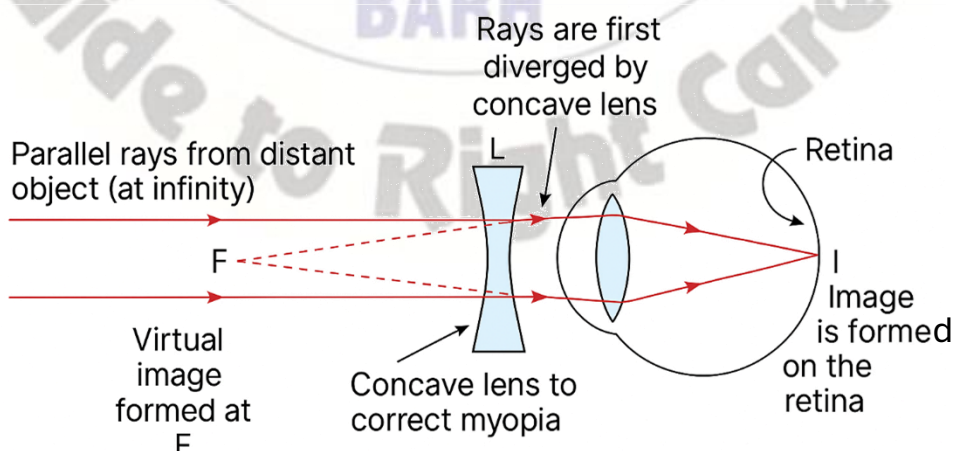
1. Myopia OR Short – sightedness OR near – sightedness

- That defect of vision in which a person can not see the distant object clearly but he can see nearby object clearly, this defect is called myopia.
- This defect arises mainly due to two reasons
 - a) Excessive curvature of the eye – lens OR High converging power of the eye – lens (because of its short focal length).
 - b) Elongation of the eye – ball OR eye – ball being too long.



Correction

- Myopia is corrected by concave lens or spectacles containing concave lenses.



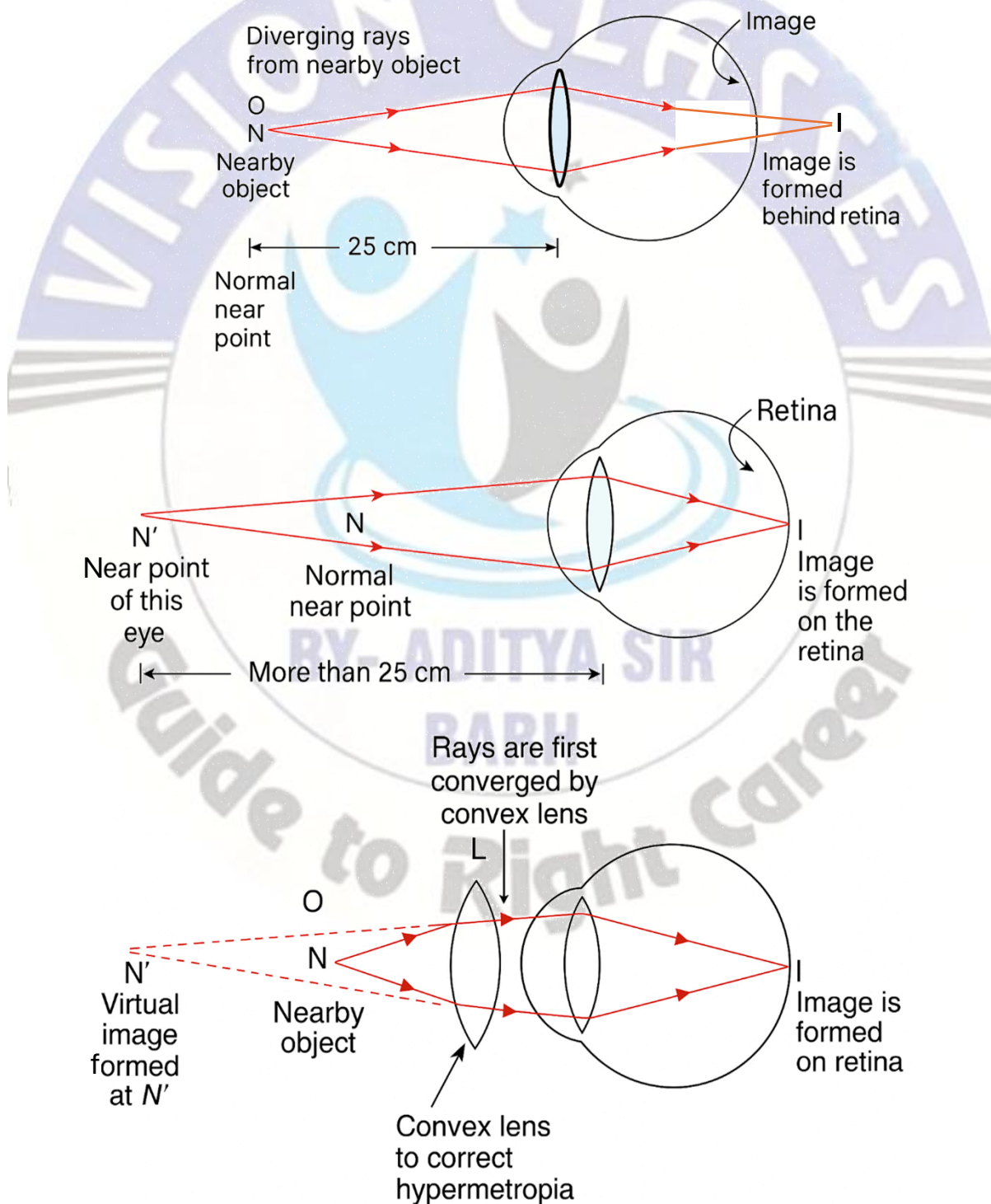
- In this case of myopia, concave lens reduces the converging power of the eye lens i.e, increases the focal length.

Note:

- The power of lens which is used in myopia is negative (-).

2. Hypermetropia OR Long – sightedness OR Far – sightedness

- That defect of vision in which a person cannot see nearby object clearly but he can see distant object. Clearly, called hypermetropia.
- The near – point of a hypermetropia eye is more than 25 cm away.
- It arises mainly due to two reasons
- a) Due to low converging power of eye – lens. Because of its large focal length.
 - b) Due to eye – ball being too short.



- In Hypermetropia, the convex lens increases the converging power of the eye lens. i.e, decreases the focal length.

Note:

- The power of lens which is used in Hypermetropia is positive (+).

3. Presbyopia

- That defect of human eyes due to which an old person can not see the nearby object clearly due to loss of power of accommodation of the eye.

Cause

- In old age, due to ciliary muscles becoming weak and the eye – lens becoming inflexible, the eye losses its power of accommodation.

Correction

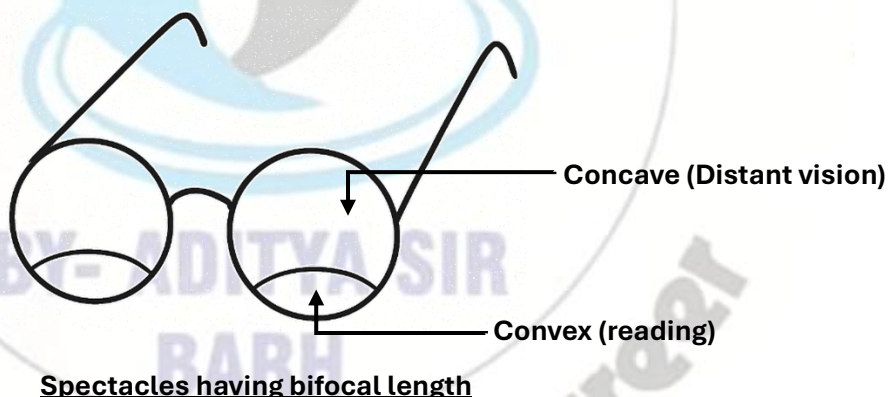
- If an old person cannot see nearby objects, then he uses only spectacles having convex lens.

Note:

- It is also possible that the same person (old) has both of defects of vision myopia as well as hypermetropia.

Correction

- If an old person cannot see far (distant) as well as nearby object then he uses spectacles having convex lens as well as concave lens.

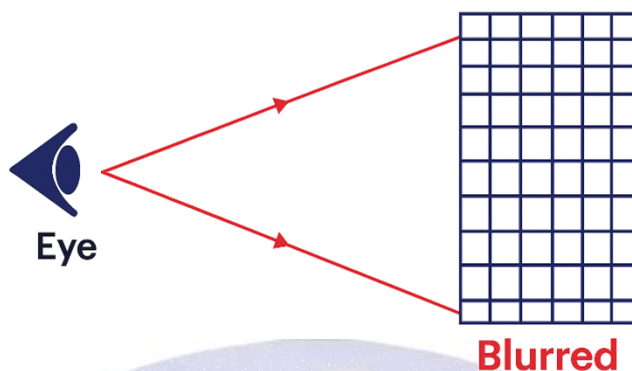


Note:

- In modern days it is possible that the refractive defects (myopia and hypermetropia) is corrected by contact lenses or, by undergoing surgical procedures. Then no need to spectacles.

- **Cataract**

- The medical condition in which the lens of the eye of a person becomes progressively cloud resulting in blurred vision, this defect is called cataract.



- This defect of vision which usually comes in old age is the cataract and person totally loses the vision of the eye.

Correction

- Cataract is only corrected by surgery, in the surgery the opaque lens is removed, and new artificial lens is inserted.

Note:

- In cataract we do not use any type of spectacles having lenses.

Q. Why do we have two eyes for vision and not just one?

Ans we have two eyes which give a wide field of view.

One eye: Horizontal field of vision about – 150°.

Two eye: Horizontal field of view about – 180°.

Having two eyes enables us to judge distances more accurately.

Q. Why a normal eye not able to see clearly the objects placed closer than 25 cm.

Ans A normal eye is not able to see clearly the objects placed closer than 25 cm because ciliary muscles cannot contract beyond a certain limit.

Q. What happens to the image distance in the eye when we increase the distance of an object from the eye?

Ans When the distance of an object from the eye is increased, the distance of image in the eye lens remains unchanged but focal length of eye lens increases so as, to form the image at same distance.

Special Note:

1. The near point (N') of hypermetropia eye is further away from the normal near point N .
i.e, $N' > N$
2. The far point (F) of a myopic eye is less than infinity.

Numerical Based Problem (NCERT)

Q1. The far point of a myopic person is 80 cm in front of the eye. What is the nature and power of lens required to correct the defect?

Ans The defect is myopia so, we used concave lens

∴ Object distance (u) = $-\infty$

∴ Image distance (v) = – 80 cm

→ Far point of eye

∴ Focal length = ?

Using lens formula

$$\Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{-80} - \frac{1}{-\infty}$$

$$\Rightarrow \frac{1}{f} = \frac{-1}{80} + 0$$

$$\Rightarrow \frac{1}{f} = \frac{-1}{80} \quad \left[\frac{1}{\infty} = 0 \right]$$

$$\Rightarrow f = -80 \text{ cm}$$

$$\begin{aligned} \therefore \text{Power of lens (P)} &= \frac{100}{f} \quad (\because f \text{ in cm}) \\ &= \frac{100}{-80} = -1.25 \text{ D} \end{aligned}$$

∴ Required power of lens (P) = -1.25 D

Q2. The near point of a hypermetropic eye is 1 m. What is the nature and power of the lens required to correct this defect?

Ans The defect is hypermetropia, so we used convex lens

∴ Object distance (u) = -25 cm

∴ Image distance (v) = -1 m
= -100 cm [near point of defect of eye]

(It is assumed that the near point of a normal eye = 25 cm)

∴ focal length (f) = ?

Using lens formula,

$$\Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{-100} - \frac{1}{-25}$$

$$\Rightarrow \frac{1}{f} = \frac{-1}{100} + \frac{1}{25} = \frac{-1+4}{100} = \frac{3}{100}$$

$$\Rightarrow \frac{1}{f} = \frac{100}{3} = 33.34 \text{ cm}$$

$$\therefore P = \frac{100}{f} = \frac{10000}{33.34} = +3\text{D}$$

∴ Required power of eye (P) = 3D

THE COLOURFUL WORLD

- Primary/Basic Colours

- The primary colours are those colours which formed a new colour when it mixed with a suitable proportion.

RGB {
1. Red
2. Green
3. Blue

- Secondary colours

- Secondary colours are those colours which formed by the mixing of primary colours.

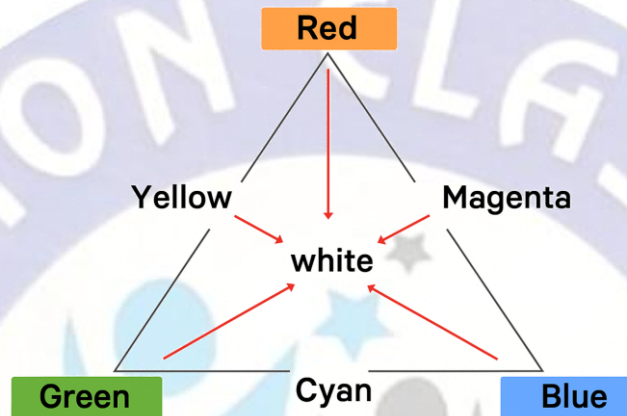
MCY {
1. Magneta
2. Cyan
3. Yellow

- **Complementary colours**

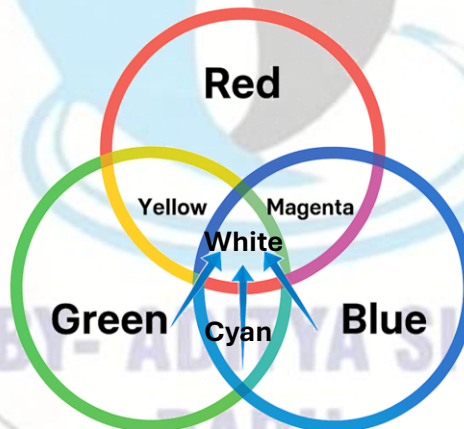
➤ Those colours which formed white colour by mixing, called complementary colours.

TRICK

1.



2.



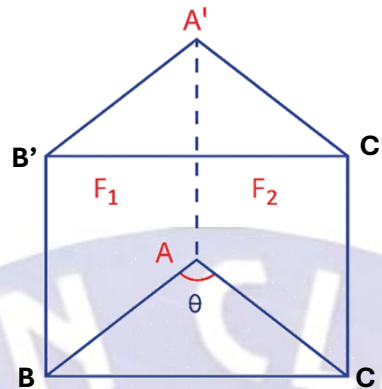
Note:

1. Red	+	Green	→	Yellow
Yellow	+	Blue	→	White
Blue	+	Green	→	Cyan
Cyan	+	Red	→	White
Red	+	Blue	→	Magenta
Magenta	+	Green	→	White

2. White colour means Sun's light.

- **Prism**

- Prism is a transparent glass having two triangular ends and three rectangular lateral faces. These faces are inclined to each other.



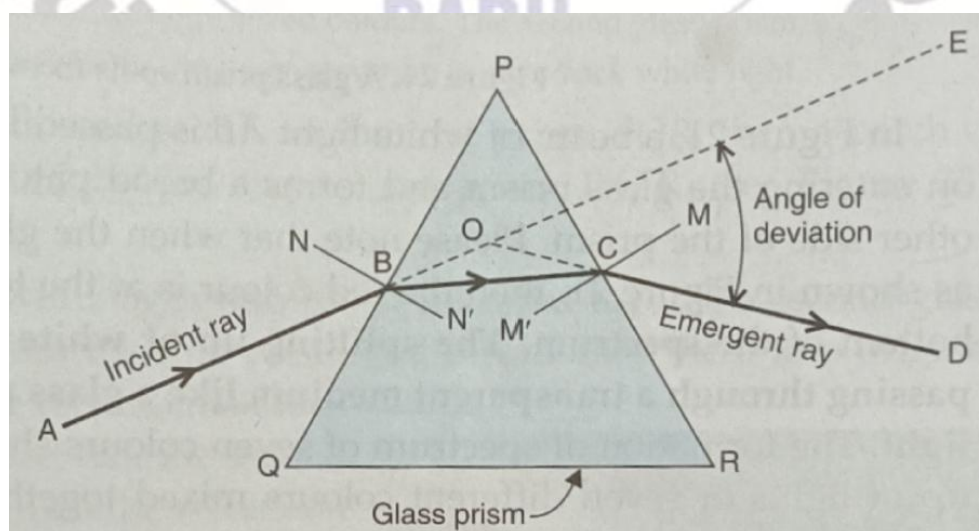
- The opposite faces F_1 and F_2 of a triangular glass prism are not parallel to each other.
- The faces F_1 and F_2 are called refracting surfaces.
- A triangular glass prism is usually called 'glass prism' OR simply called prism.
- The line where faces are joined is called refracting edge; (AA')
 $\therefore \angle CAB = 60^\circ$
- All rectangular faces are inclined at angle at 60° to one – another.
- The angles between the opposite faces are called the angle of the prism.
- ABC is called principal section of prism.
- The angle of principal section is 60° . So, it is equilateral triangle. i.e, The base of a prism is an equilateral triangle.
- Total no. of faces in a prism = 5.

- **Refraction of light through a glass prism**

- When a ray of light passes through a glass prism, refraction of light occurs both, when it enters the prism as well as when it leaves the prism. Since the refracting surface of prism are not parallel, therefore the emergent ray and incident ray are not parallel to one another. In this case the ray of light is deviated on passing through the prism.

- **Angle of deviation**

- When a ray of light passes through a prism, it bends towards the base of prism then the angle between incident ray and emergent ray is called angle of deviation.



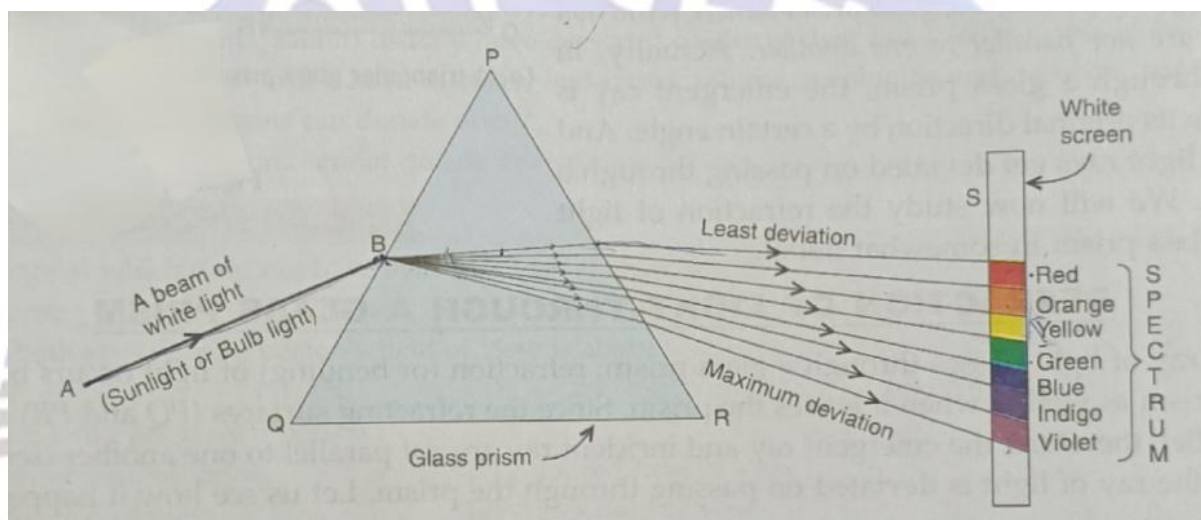
- The angle of deviation is denoted by δ (Delta).

▪ **Angle of deviation depends on**

- Angle of deviation (δ) $\propto$ Angle of incidence (i).
- Angle of deviation (δ) $\propto$ refractive index (μ) of glass prism.
- Angle of deviation (δ) $\propto$ Angle of glass prism.
- Angle of deviation (δ) $\propto \frac{1}{\text{Wavelength of light}}$.

• **Dispersion of Light (Sun light)**

- When the sun's light ray passes through a glass prism, it splits into seven colours on the other side of prism. This phenomenon is called dispersion of light.
- OR**
- The split up of white light into seven colours on passing through a glass prism is called dispersion of light.



- The above experiment, shows that the sun light is mixture of seven colours.
- The dispersion of white light into seven colours only possible because different colours of light has different wavelength and speed.
- The wavelength of light $\propto$ speed of light.
i.e, Higher the wavelength, higher the speed of light.

Note:

- The wavelength of red colour is maximum. So, the speed of red colour is maximum but deviation is least.
i.e, $\lambda (\uparrow)$, speed $(\uparrow)$, $\delta (\downarrow)$
- The wavelength of violet colours is minimum. So the speed minimum but deviation is maximum.
i.e, $\lambda (\downarrow)$, speed $(\downarrow)$, $\delta (\uparrow)$

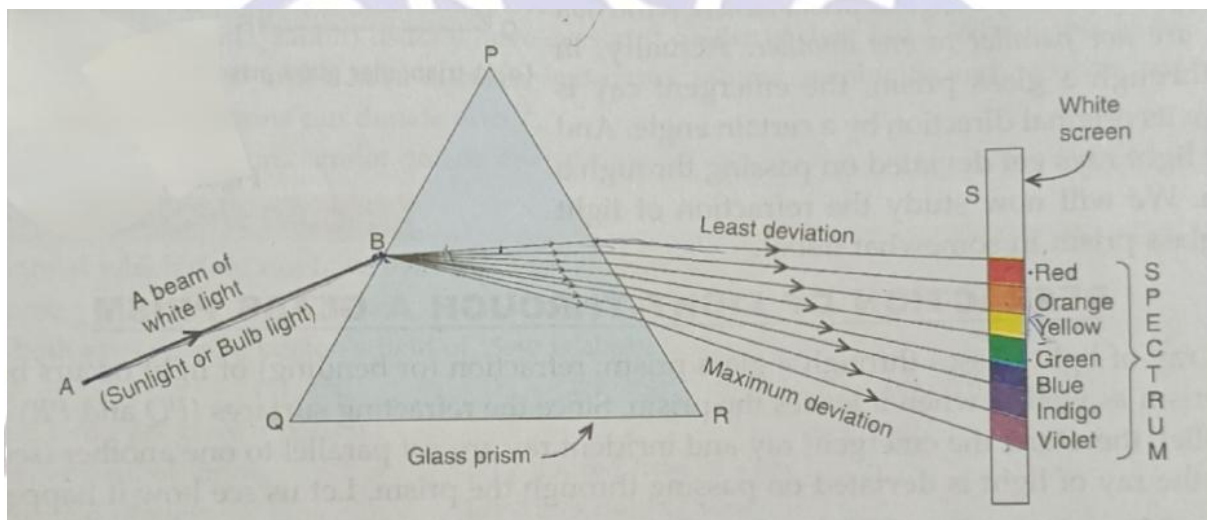
Colour	Wavelength (in nm)	Deviation
Red	620 – 700	Minimum  Maximum
Orange	590 – 620	
Yellow	570 – 590	
Green	500 – 570	
Blue	460 – 500	
Indigo	440 – 460	
Violet	400 – 440	

TRICK

V	I	B	G	Y	O	R
↓	↓	↓	↓	↓	↓	↓
Violet	Indigo	Blue	Green	Yellow	Orange	Red

- **Define Spectrum with diagram**

- When the ray of sun light passes through a glass prism then the sun light splits to form a band of seven colours on a white screen, this screen is known as spectrum or, spectrum of white light.

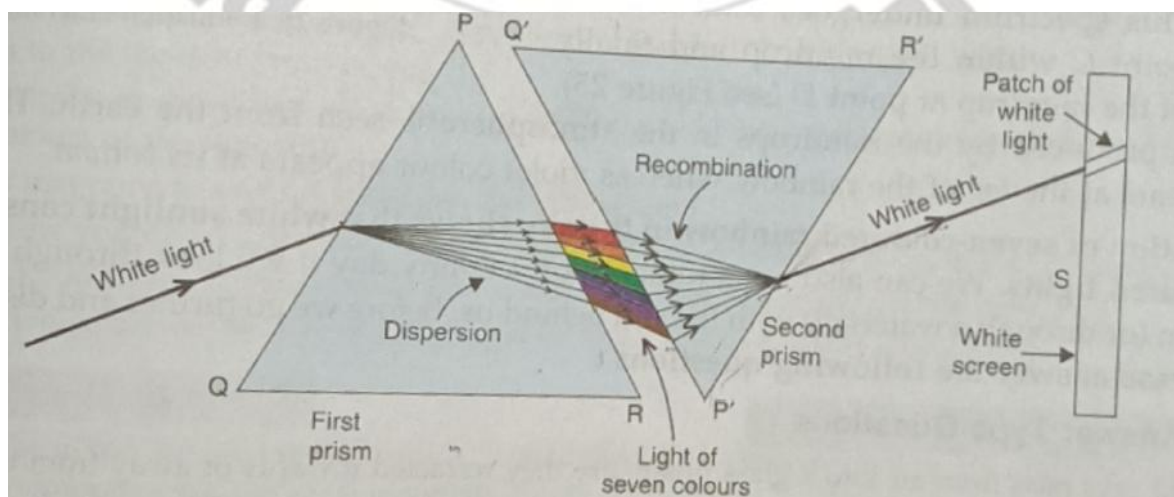


Note:

- The speed of any colour is maximum in the prism then its deviation is minimum and its vice – versa.

- **Re – combination of spectrum colours to give white light**

- The components of light separated out by a prism can be re combined by passing then through a prism kept inverted respect to the first then second glass prism recombined the seven colours of spectrum to give back white light.



- **The rainbow**

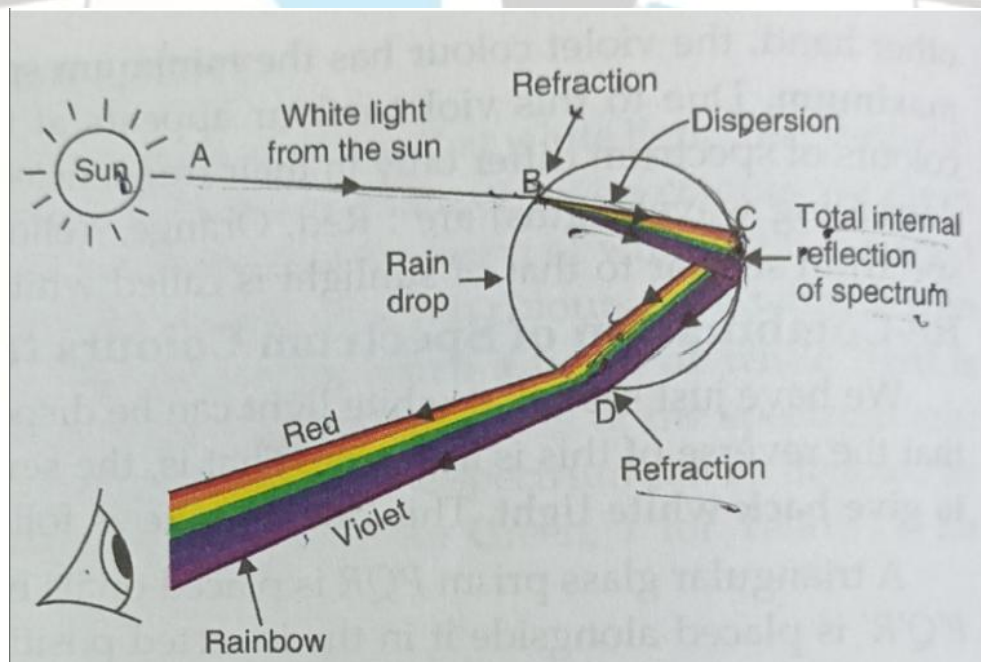
- The rainbow is perhaps the most beautiful optical phenomenon. It is an arc of seven colours visible in the sky when there are raindrops in the air.
- It is visible when the sun is low in the sky (early morning and late afternoons) and it is raining at the same time.
- The rainbow is actually a natural spectrum of sunlight in the sky.
- The rainbow can be seen when we stand with our back towards the sun and rain in front of us.
- A rainbow is always formed in the direction opposite to that of the sun.



Rainbow

- **Formation of rainbow**

- A rainbow is produced by the dispersion of white sunlight by raindrops in the atmosphere. Each raindrop acts as a tiny glass prism splitting the sunlight into a spectrum visible in the sky.
- It is visible when the sun is low in the sky. i.e, early morning and evening.
- The rain drops in the atmosphere act like many small prism.



- **Atmospheric refraction**

- In the atmosphere air having different layers and each layer has different optical densities, the density of cold air is high (denser) while the density of warm air is low density (rarer).

The refraction of light caused by the earth's atmosphere is called atmospheric refraction.

Due to atmospheric refraction

- a) The twinkling the stars.

- b) The stars seem higher than they actually are.
- c) Advance sunrise and delay sunset etc.

- **Twinkling of stars OR stars seems to twinkle why?**

- The distance of stars from the earth is very large, The continuously changing atmosphere refracts the light when the atmosphere refracts more, star – light towards us, the star appears to be bright and when the atmosphere refracts less star – light then the stars appear to be dim.

In this way star – light reaching our eyes increases and decreases continuously due to atmospheric refraction and the stars appear to twinkle at night.

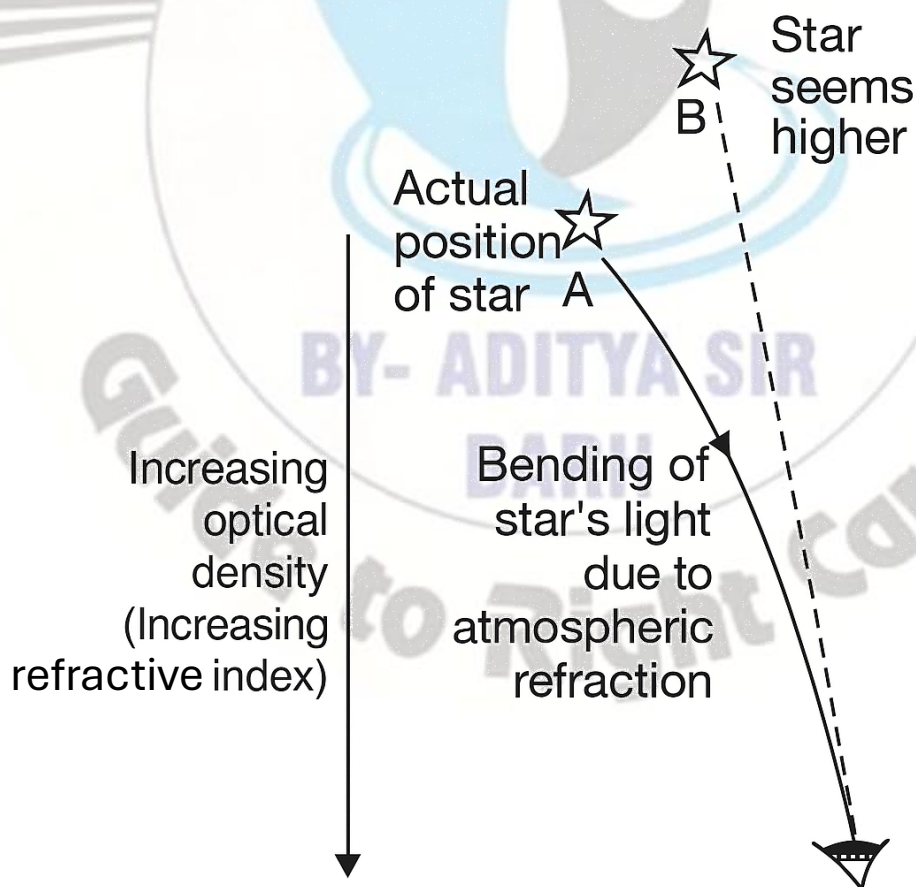
Special Note:

- Stars twinkle at night but planets do not twinkle because planets are big in size and they are much nearer to the earth and planets have no own light.

- **The stars seem higher than they actually are**

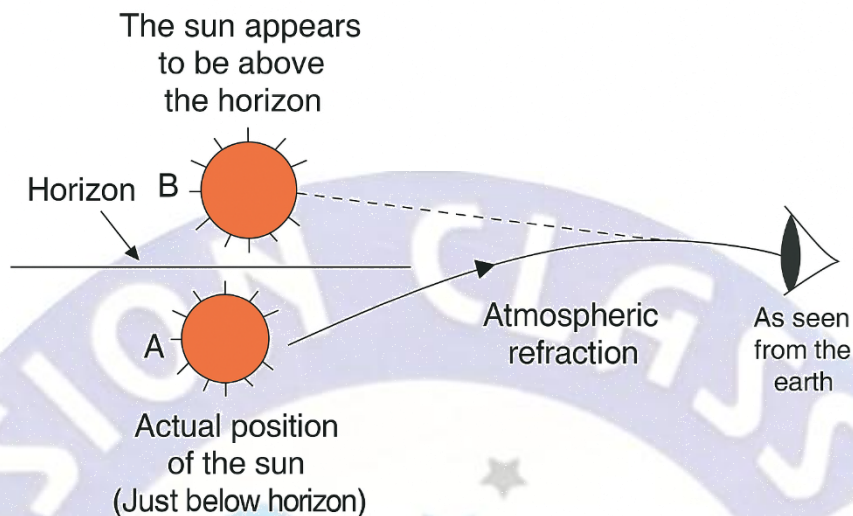
- Due to atmospheric refraction, the stars seem to be higher in the sky than they actually. This can be explained as follows:

Light from a star is bent (refracted) as it leaves space (a vacuum) and enters the earth's atmosphere. Air higher up in the sky is rarer but that nearer the earth's surface is denser. So, as the light from a star comes down, the dense air bends the light more. Due to this refraction of star's light, the star appears to be at a higher position.



- **Advanced Sunrise and Delayed Sunset**

- We can see the sun about 2 minutes before the actual sunrise and 2 minutes after the actual sunset because of atmospheric refraction.

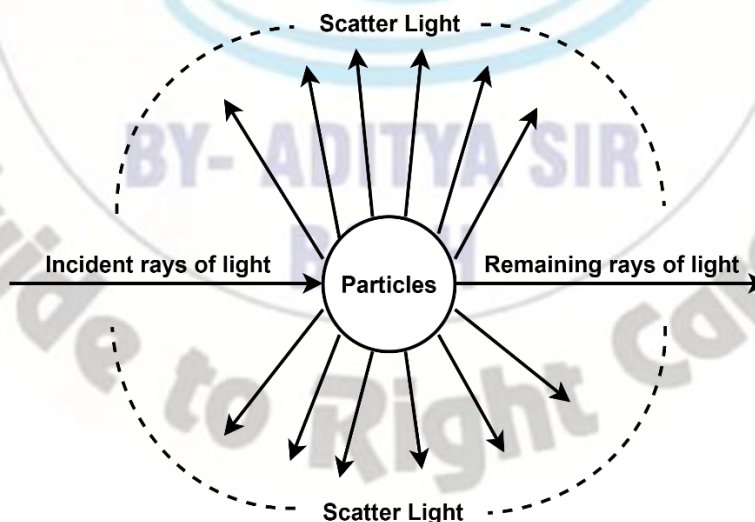


Note:

- The time from sunrise to sunset is lengthened by about $2 + 2 = 4$ minutes because of atmospheric refraction.

- **Scattering of light**

- The phenomenon in which a part of the light incidents on a particle is called scattering of light.



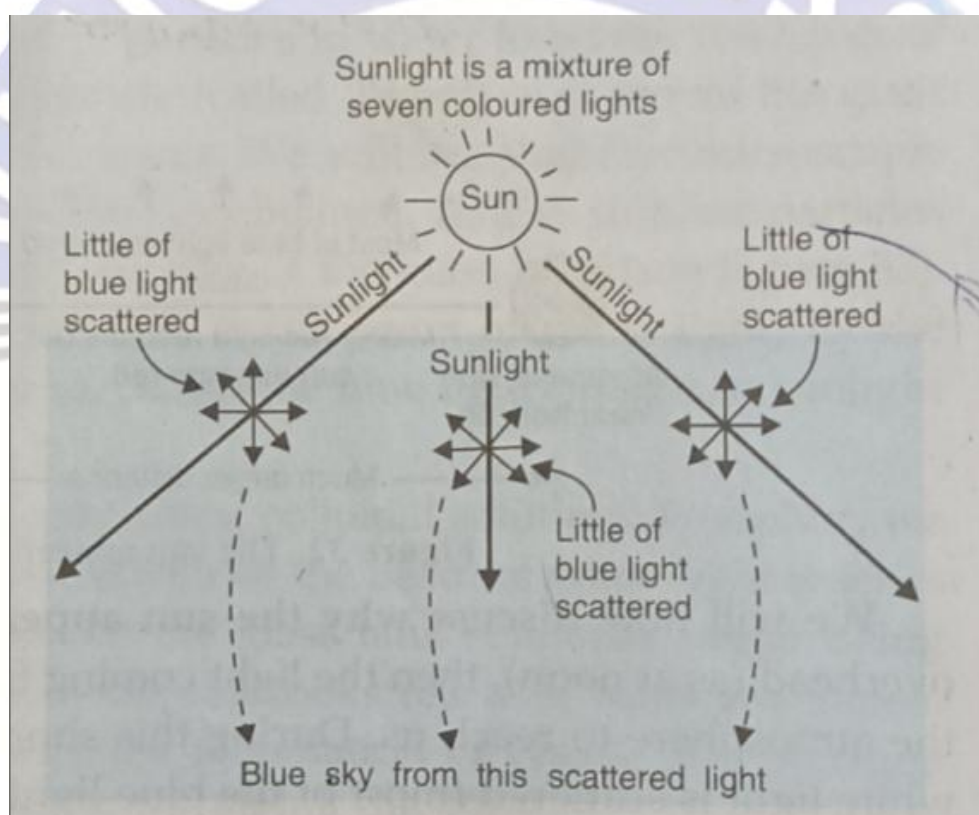
Examples of scattering of light

- ✓ The sky's colour appears blue.
- ✓ The sunset and sunrise are red.
- ✓ The scattering of sun light passing through the canopy of a forest.
- ✓ Tyndall effect.

- **The colour of scattered light depends on the size of the particles**
 - a) Dust particles and water droplets suspended in the atmosphere are much larger than the wavelength range of visible light.
 - b) The air molecules like as: N_2 , O_2 gas molecules present in the atmosphere are smaller than the wavelength range of visible light.
- **Conclusion from above result**
 - The colour of the scattered light depends on the size of the scattering particles in the atmosphere.
 - a) The larger particles scattered the white light.
 - b) The smaller particle scattered the blue light.

Q. Why does sky seems blue?

Ans When the sunlight enters in earth's atmosphere, the scattering of blue components of white sunlight by air molecules present in the atmosphere causes the blue of the sky.



Note:

- If the earth has no atmosphere then no scattering of sunlight and skyworld have looked dark and black to us.

Q. Why does the sky appear dark instead of blue to an astronaut?

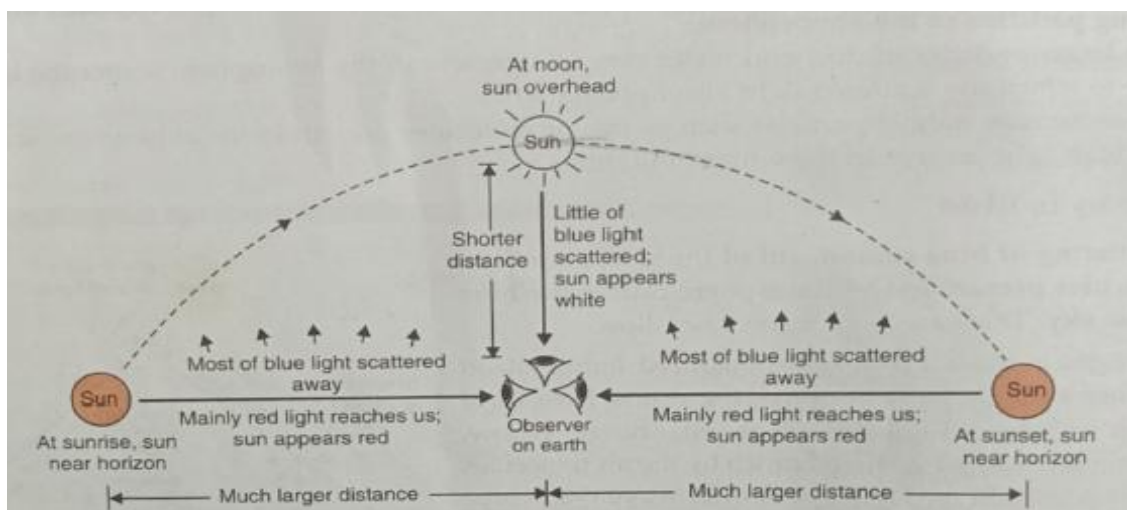
Ans An astronaut is much above the earth's atmosphere in the absence of air atmosphere there is no scattering of light as a result, so the sky appears dark to an astronaut.

Q. Explain why the planets and moon do not twinkle?

Ans Planets and moon are larger in size and closer to the earth. The total amount of reflected light entering our eye from a planet is comparatively large as a result, In spite of atmospheric refraction the total quantity of light entering the eye of observer from a planet remains constant. So the planets do not twinkle.

Q. Explain why the sun appears red at sunrise and sunset?

Ans Because at the time of sunrise and sunset sun is near the horizon and light from the sun passes through a larger distance in the earth's atmosphere. During this passing light of shorter wavelengths is mostly scattered and the light reaching the eye of observer is rich in longer wavelengths. So the sun appears to be reddish/red.



Note:

- The sun appears white when it is overhead in the sky because it travel shorter distance to the atmosphere to reaching us.

Q. Why do we use red colour in signals of railway, airplane, T.V tower, radio tower, telephone tower etc.

Ans The red colour light used in signals because it has longer – wavelength in visible light, so the red colour is scattered the least by air molecules of the atmosphere and therefore it can reach to a longer distance.

Note:

- Astigmatism arises due to irregular shape of cornea or lens.
- In Astigmatism a person has blurred vision.
- Astigmatism is corrected by cylindrical lens.